Lab Log Book

WEEK - 5(Jagadeesh)

1. Model Summary

The following screenshot shows the architecture of my modified CNN model with convolutional kernel size = 5.
The model contains Conv1D, MaxPooling, GlobalMaxPooling, and Dense layers.
The total number of trainable parameters is 28,977.

```
from tensorflow import keras
import tensorflow as tf

model = keras.Sequential([
    keras.layers.Conv1D(50, 5, padding='same', input_shape=(50, 5), activation=tf.nn.relu, kernel_initializer="normal"),
    keras.layers.MaxPooling1D(7),
    keras.layers.Conv1D(100, 5, padding='same', activation=tf.nn.relu, kernel_initializer="normal"),
    keras.layers.GlobalMaxPooling1D(),
    keras.layers.Dense(25, activation=tf.nn.relu, kernel_initializer="normal"),
    keras.layers.Dense(2)
])

model.summary()
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
conv1d_2 (Conv1D)	(None, 50, 50)	1,300
max_pooling1d_1 (MaxPooling1D)	(None, 7, 50)	0
conv1d_3 (Conv1D)	(None, 7, 100)	25,100
global_max_pooling1d_1 (GlobalMaxPooling1D)	(None, 100)	0
dense_2 (Dense)	(None, 25)	2,525
dense_3 (Dense)	(None, 2)	52

Total params: 28,977 (113.19 KB)
Trainable params: 28,977 (113.19 KB)
Non-trainable params: 0 (0.00 B)

2. Model Training

```
history = model.fit(X_train, y_train, batch_size=50, epochs=epochs, validation_split=0.2, verbose=1)

mse, mae = model.evaluate(X_test, y_test, verbose=1)
print("Test MAE:", mae)
print("Test MSE:", mse)

pred = model.predict(X_test, verbose=0)
```

```
Epoch 1/17
3520/3520 ━━━━━━━━━━━━━━━━━━━━ 7s 2ms/step - loss: 0.0021 - mae: 0.0268 - val_loss: 9.3127e-04 - val_mae: 0.0202
Epoch 2/17
3520/3520 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 7.6642e-04 - mae: 0.0189 - val_loss: 8.6853e-04 - val_mae: 0.0190
Epoch 3/17
3520/3520 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 7.2232e-04 - mae: 0.0181 - val_loss: 8.5910e-04 - val_mae: 0.0188
Epoch 4/17
3520/3520 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 7.0982e-04 - mae: 0.0179 - val_loss: 8.5372e-04 - val_mae: 0.0189
Epoch 5/17
3520/3520 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 7.0370e-04 - mae: 0.0178 - val_loss: 8.5810e-04 - val_mae: 0.0191
Epoch 6/17
3520/3520 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 6.9950e-04 - mae: 0.0177 - val_loss: 8.4938e-04 - val_mae: 0.0188
Epoch 7/17
3520/3520 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 6.9686e-04 - mae: 0.0177 - val_loss: 8.4010e-04 - val_mae: 0.0189
Epoch 8/17
3520/3520 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 6.9065e-04 - mae: 0.0175 - val_loss: 8.3348e-04 - val_mae: 0.0187
Epoch 9/17
3520/3520 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 6.8913e-04 - mae: 0.0175 - val_loss: 8.2597e-04 - val_mae: 0.0184
Epoch 10/17
3520/3520 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 6.8811e-04 - mae: 0.0175 - val_loss: 8.7509e-04 - val_mae: 0.0195
Epoch 11/17
3520/3520 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 6.8653e-04 - mae: 0.0175 - val_loss: 8.2021e-04 - val_mae: 0.0184
Epoch 12/17
3520/3520 ━━━━━━━━━━━━━━━━━━━━ 7s 2ms/step - loss: 6.8111e-04 - mae: 0.0174 - val_loss: 8.2327e-04 - val_mae: 0.0187
Epoch 13/17
3520/3520 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 6.8016e-04 - mae: 0.0174 - val_loss: 8.3253e-04 - val_mae: 0.0187
Epoch 14/17
3520/3520 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 6.7812e-04 - mae: 0.0173 - val_loss: 8.6146e-04 - val_mae: 0.0194
Epoch 15/17
3520/3520 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 6.7844e-04 - mae: 0.0173 - val_loss: 8.3696e-04 - val_mae: 0.0186
Epoch 16/17
3520/3520 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 6.7997e-04 - mae: 0.0173 - val_loss: 8.2143e-04 - val_mae: 0.0185
Epoch 17/17
3520/3520 ━━━━━━━━━━━━━━━━━━━━ 6s 2ms/step - loss: 6.8005e-04 - mae: 0.0174 - val_loss: 8.5202e-04 - val_mae: 0.0192
936/936 ━━━━━━━━━━━━━━━━━━━━ 0s 434us/step - loss: 6.2064e-04 - mae: 0.0169
```

```
Test MAE: 0.016884515061974525
Test MSE: 0.0006206360994838178
```